Continuous Distillation

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Introduction

- **The purpose of this experiment is to examine the effects of varying feed position under continuous operation in distillation**
- **<u>Components</u>:** Refractive Index, Composition, Mole Fractions, Relative Volatility, Overall Column Efficiency
- **<u>Parameters</u>:** Feed Positions (Midpoint and Top)
- **<u>Key</u>:** Take distillate and bottom samples at same time, ensure correct calibration curve is done in order to deliver accurate flow rate in column

Theory

- *McCabe-Thiele*
  - $y_{n+1} = \frac{R}{(R+1)} X_n + \frac{1}{(R+1)} X_D$
- *Column Efficiency*
  - $\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}}$
- *Relative Volatility*
  - $\alpha = \frac{(\frac{y_i}{x_i})}{(\frac{y_j}{x_j})} = \frac{K_i}{K_j}$
- *Fenske Equation*
  - $N = \frac{log\left[\left(\frac{X_d}{1-X_d}\right)\left(\frac{1-X_b}{X_b}\right)\right]}{\log(\alpha)}$

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Apparatus

Number	Instrument	Description
1	Distillation Column	Glass distillation column with sieve trays where the VLE is obtained; Separates 10 wt% ethanol-water solution
2	Condenser	Coiled glass tubing where the cooling water flows; Cools the vapor coming out of the distillation column
3	Reboiler	Contains the feed (10 wt% ethanol) and boils feed into the distillation column
4	Heater	Used to heat the reboiler
5	Pump	Pushes feed back into reboiler
6	Funnel to Bottom	Used to charge the reboiler at the start of the experiment
7	Bottom Product Receiver	Collects the bottom product from the reboiler
8	Top Product Receiver	Collects the top product from the decanter
9	Reflux Valve	Controls the reflux within the column
10	Decanter	Decants the product coming out of the condenser into the top product tank
11	Manometer	Used to determine the pressure drop across the column
12	Control Panel	Used to manipulate the automatic valves in the process
13-20	Thermocouple 1 - Thermocouple 8	Monitors the temperature of the respective stage in the column
21	Valve 6	Located on cooling water pipe to control flow
22	Valve 7	Controls the pressure reading within the column
23	Valve 11	Controls the cooling water entering the condenser
24	Valve 5	Three-way valve that collects sample, closes reboiler, or allows for sample collection
25	Rotameter (not shown in diagram)	Used to measure the flow of cooling water through condenser
26	Temperature gauge (not shown in diagram)	Used to measure the temperature of the cooling water
27	Glass Jar	To hold the feed liquid being pumped into the column

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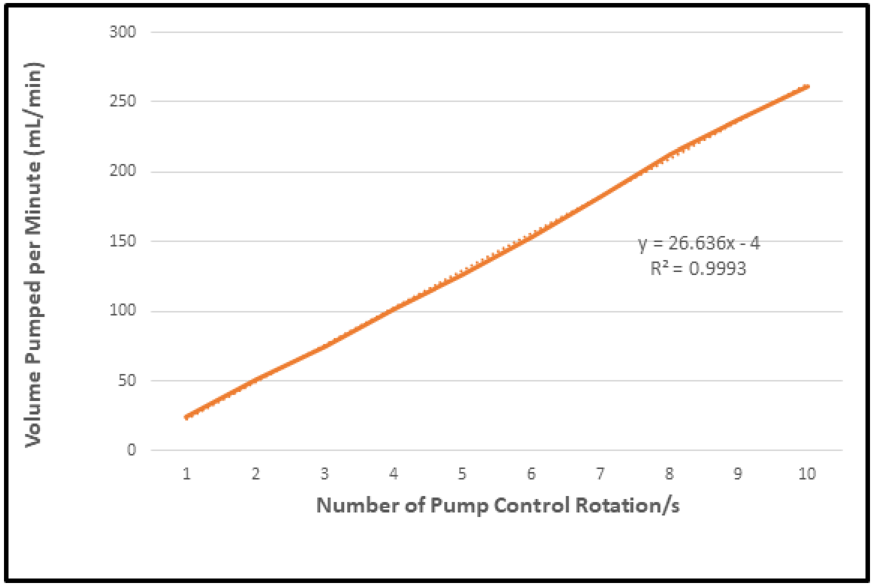
Material & Supplies

- Ethanol
- DI Water
- Refractometer
- Test Tubes
- Kimwipes
- Cold Water Bath/Ice
- Heat Resistant Gloves
- Wrench

Pump Calibration Procedure

Experiment A: Speed Pump Calibration

- Fill glass jar with 3000 ml of DI water.
- Disconnect the feed tube from column.
- Pump speed will be varying from 1-10 on the dial (interval of 1).
- Allow for the flow to settle.
- Measure the amount of fluid that is expelled from the tube in 10 seconds.
- Plot the data to obtain the dial number needed to push 30 ml/min into system



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Procedure (Continued)

- Part b: Using Midpoint Feed Position
- Add 10wt% EtOH solution in reboiler tank, and 30wt% EtOH in the feed glass vial
- Open yellow cooling water valve and set to 3L/min, set power to 1.5kW until at steady state. Then reduce to 0.75kW.
- Turn on feed pump to correct calibration in order to feed 30-35mL/min. Wait until 500mL of feed is sent into column before sampling.
- Each trial will require you to take sample from top and bottom at the same time. Use these obtained samples to run the Refractive Index, (3 sets of samples, 5 minute interval)
- Part C: Above Top Plate Feed Position
- Prior to starting part C, turn off feed pump, reboiler power, and close valve V1.
- Use wrench and professor’s assistance in setting up top feed position. Perform same procedure as above part B.

Experimental Challenges

- Pre-Calculate solution of 10wt%, 30wt% EtOH
- Make solution as early as possible to start boiler ~ 30-45min
- Don't forget to open valve for cooling water
- Zero the RI before each use
- Ensure pump suction tube isn't at the bottom of the vial

Safety

- Ensure pump, reboiler power, and reflux switch are off when changing feed positions. Also, ensure valve V1 is closed
- Samples, especially those taken from the reboiler (still), will be hot. Use insulated gloves when taking those samples.
- Wear gloves and goggles at all times.